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Journal of Manufacturing Processes

Date: 30 October 2025

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Process monitoring via prediction of laser beam absorptance in partial penetration laser keyhole welding of aluminum alloy

[Kimoon Nam](#), [Hyungson Ki](#)   [Show more](#) ▼[View PDF](#)[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.jmapro.2025.07.088](#) ↗[Article](#)[Recommended articles](#)[Cited by \(2\)](#)[Metrics](#)

Abstract

Achieving high-quality laser keyhole welding requires precise control of energy absorption, which is inherently linked to the complex shape and dynamics of the keyhole. While absorptance-based monitoring methods have been successfully applied to full penetration welding, their application to partial penetration welding remains a challenge due to the limited visibility of the keyhole interior. In this study, we developed a deep learning-based absorptance monitoring system for partial penetration laser welding, utilizing a combination of object detection and CNN regressors. The system estimates keyhole aperture shape and penetration depth from melt pool images and constructs a two-channel keyhole representation to predict laser absorptance. The proposed method was trained and validated using melt pool image datasets collected from 14 fiber laser

Outline

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welding experiments on aluminum alloy 1050P-H16, covering laser power levels (538–739W) and scanning speeds (8.2–15.4mm/s). The deep learning model achieved high accuracy in predicting laser beam absorptance, with an R-squared value of 0.99706, closely matching ray tracing simulations. Additionally, three distinct absorptance patterns were identified, correlating with variations in penetration depth, weld bead shape, and defect formation. This study presents the first application of deep learning for absorptance prediction in partial penetration keyhole welding. The proposed framework enables real-time welding quality assessment and provides insight into hidden keyhole dynamics, addressing a critical gap in laser welding research.

Keywords

Deep learning; Laser welding monitoring; Partial penetration; Laser beam absorptance; Aluminum alloy

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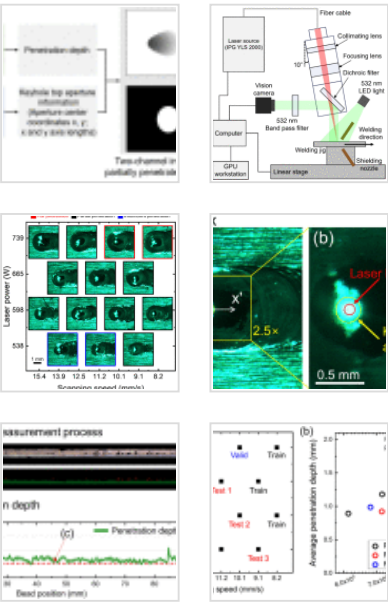
1. Introduction

The success of any welding process hinges on delivering the precise amount of energy. This is especially true for laser keyhole welding, where the absorption of energy is intricately linked to the complex shape and dynamics of the keyhole itself [1,2]. Recognizing this connection, researchers have devoted significant effort to studying the relationship between keyhole shape and energy absorption, aiming to gain a deeper understanding and improve control over laser welding processes.

Kawahito et al. [3] employed in situ x-ray transmission observation to analyze the keyhole geometry in 304 stainless steel and A5052 aluminum alloy. Additionally, they determined the laser beam absorption by measuring the temperature variation in the

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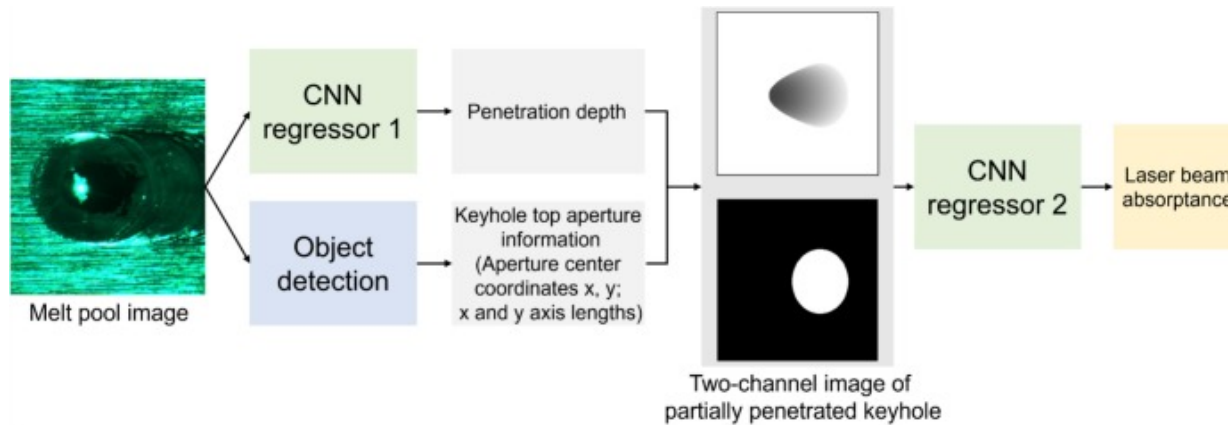
water surrounding the specimen surface during laser irradiation. Their findings revealed a linear relationship wherein an increase in laser input energy corresponds to a deeper keyhole, resulting in enhanced absorptance. Martin et al. [4] conducted an experiment utilizing the sandwich method and observed keyholes in aluminum (Al6061) and titanium (Ti-6Al-4V) through transmission x-ray imaging. They also conducted multiphysics simulations of laser-metal interactions using a ray-tracing method. In their experiments, they observed an expansion of the keyhole's rear wall, which was identified as an instability caused by the recoil pressure. Gan et al. [5] observed keyholes in aluminum (Al6061), stainless steel (SS316), and titanium (Ti-6Al-4V) using high-speed x-ray imaging. They compared the photographed keyholes to those simulated using multiphysics modeling and found a high degree of similarity. They noted that when the peak temperature exceeds the boiling point, absorption increases rapidly. This causes the keyhole to deepen, leading to changes in absorption. Fetzer et al. [6] conducted x-ray videography on the AA6082 aluminum alloy to observe keyhole formations. They directly utilized the keyhole shapes captured by x-ray imaging for ray tracing simulations. It was observed that as the scanning speed increased, the keyhole shape became shallower and elongated, leading to a gradual reduction in absorptance.

These studies revealed a strong correlation between the shape of the keyhole, the laser absorption pattern, and welding results. Building on these observations, researchers have recently developed welding monitoring systems based on laser beam absorptance for full-penetration laser keyhole welding of steel and aluminum alloys. Kim et al. [7] proposed a laser beam absorptance monitoring system for A366 steel. The system utilized a deep learning-based object detection model to identify keyhole apertures and a convolutional neural network (CNN) regressor [8] to predict absorptance. They observed a rapid increase in absorptance where the bead width narrows significantly. Kim et al. [9] further developed a laser-beam absorptance monitoring system for aluminum alloy. They classified three absorption patterns based on the multiple reflection patterns inside the keyhole and the behavior of the keyhole, and these patterns matched the bead geometry.

Nam and Ki [10] employed the absorptance monitoring system to detect burn-through defects during laser welding.

These studies have shown that absorptance prediction-based monitoring methods hold significant promise. However, their application has been limited to full-penetration laser welding. This limitation arises from the scarcity of information about the keyhole in partial penetration welding, making absorptance prediction very challenging. While the shape of a fully penetrated keyhole can be estimated from the shapes of the upper and lower apertures, performing such a task is largely impossible for partially penetrated keyholes. Given the widespread industrial applications of partial penetration laser welding, addressing this issue is important. For secondary battery manufacturing, for example, partial penetration keyhole welding is frequently used to prevent the risk of splatter ejecting into the cell and damaging the separator [11].

In this study, we developed a laser beam absorptance monitoring method for partial penetration keyhole welding processes using a deep learning approach. The overall structure of the algorithm is presented in Fig. 1. In this method, the shape of a partially penetrated keyhole was estimated from the keyhole top aperture shape and the penetration depth, both of which were predicted from a melt pool image by an object detection algorithm (YOLOv7 [14], Object detection in Fig. 1) and a CNN regressor (VoVNet27-slim [15], CNN regressor 1 in Fig. 1), respectively. The estimated keyhole shape was constructed as a two-channel input image for another CNN regressor (VoVNet27-slim, CNN regressor 2 in Fig. 1), which was trained with ray tracing simulation data and predict laser beam absorptance. Using the monitoring system, we were able to detect underfill defects and regions where weld depth changes abruptly. Predicting partially penetrated keyholes presents significant challenges due to the limitations of optical camera observation. Real-time calculation of absorptance using numerical simulations is also computationally expensive. We believe that the proposed deep learning-based method can serve as a practical and reasonable way to understand the energy transfer process inside a hidden keyhole.



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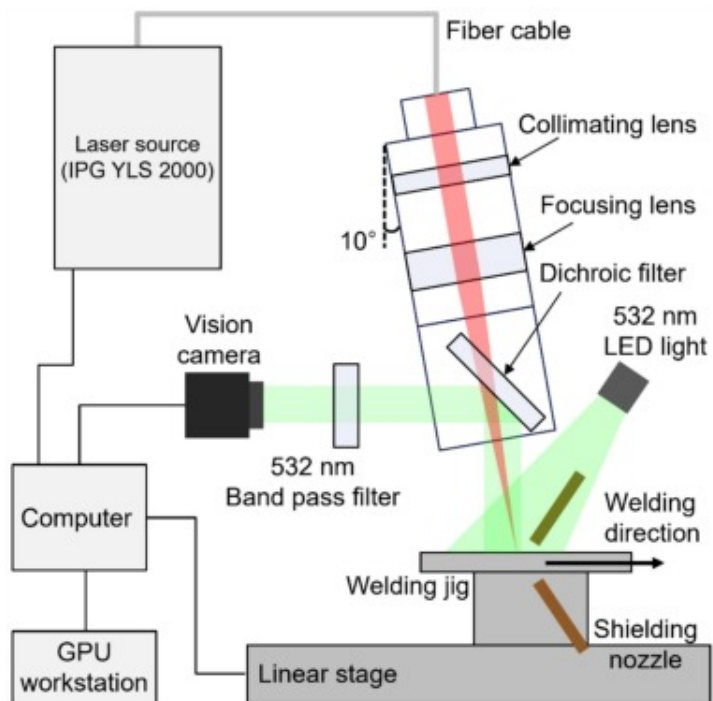
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Fig. 1. Overview of the proposed laser beam absorptance monitoring system for partial penetration keyhole welding. A melt pool image is processed using an object detection model (YOLOv7) to extract keyhole top aperture information and a CNN regressor (VoVNet27-slim) to predict penetration depth. These predictions are used to generate a two-channel image of the partially penetrated keyhole, which serves as input for a second CNN regressor trained on ray-tracing simulation data to predict laser beam absorptance.

2. Experiment

Fig. 2 illustrates the experimental setup used in this study. Welding was performed using a 2-kW continuous-wave multi-mode fiber laser (IPG YLS 2000) with a wavelength of 1070nm. The beam from the laser source was delivered through a 100 μ m fiber, then collimated by a collimating lens with a focal length of 160mm, and subsequently focused on the specimen's surface using a focusing lens with a focal length of 160mm. The focused beam had a diameter of 100 μ m with a top-hat energy distribution. The laser head was rotated 10° to avoid optical damage from reflected light. The specimen material was Al 1050P-H16, and its chemical composition is shown in Table 1. Two specimens, each with dimensions of 137mm $\times$ 30mm $\times$ 1mm, were stacked and welded together without a

gap. The weld length along the centerline of the specimen was set to 90mm. To prevent oxidation, argon shield gas was supplied at the top and bottom surfaces of the specimen at a flow rate of 15L/min.



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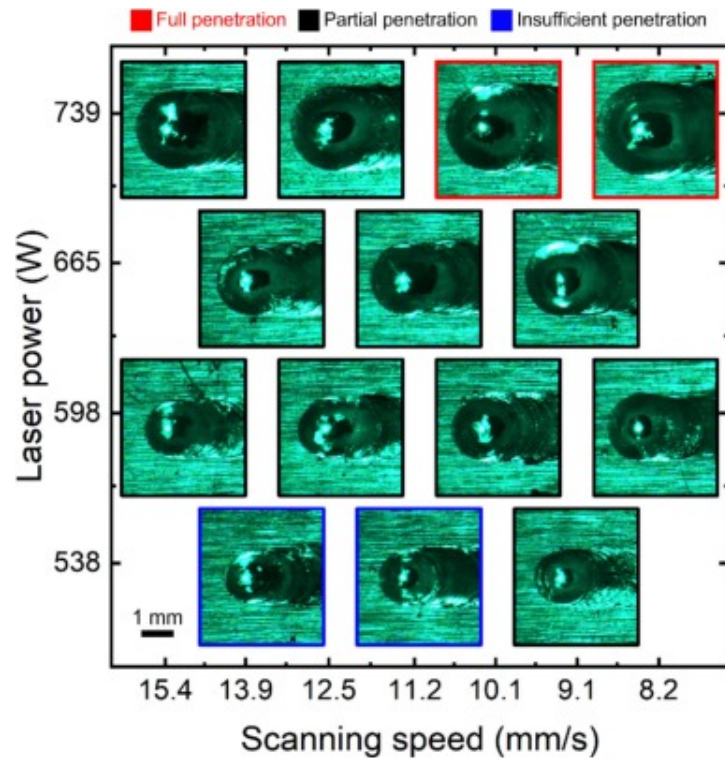
Fig. 2. Schematic of the experimental setup.

Table 1. Chemical composition (wt%) of Al 1050P-H16.

Si	Fe	Cu	Mn	V	Ti	Al
0.137	0.255	0.006	0.002	0.016	0.021	Balance

The welding process was monitored using a coaxial observation method, as shown in [Fig. 2](#). This method, detailed in [\[16\]](#), enabled us to obtain clear and distortion-free images. Additionally, we used a small optical camera (FLIR Blackfly, BFS-U3-04S2C), which offered a cost advantage over high-speed cameras, as in our previous research [\[17\]](#). A 50W 532nm light-emitting diode (LED) light was directed towards the specimen's surface, as displayed in [Fig. 2](#). The dichroic filter, installed inside the laser head at 45° angle relative to the specimen surface, transmitted the laser beam (1070nm) while reflecting the LED light (532nm). The LED light, after being reflected by the dichroic filter and passing through the bandpass filter, was detected by the camera sensor. The resolution of the image was set to 352×320pixels (4.12mm×3.74mm). The entire welding process (90mm) was captured at a rate of 700 frames per second (fps).

Fourteen experiments were conducted to collect the dataset. Four laser powers (ranging from 538W to 739W) and seven scanning speeds (ranging from 8.2mm/s to 15.4mm/s) were selected as process conditions. [Fig. 3](#) displays a process map with images representing each process condition. The process map contains 2 full penetration welds (penetration depth>2mm), 10 partial penetration welds (1mm<penetration depth≤2mm), and 2 welds of insufficient penetration (penetration depth≤1mm), which are marked with red, black and blue boxes, respectively. Since the thickness of the specimen was 1mm, joining does not occur if the penetration depth is less than 1mm.



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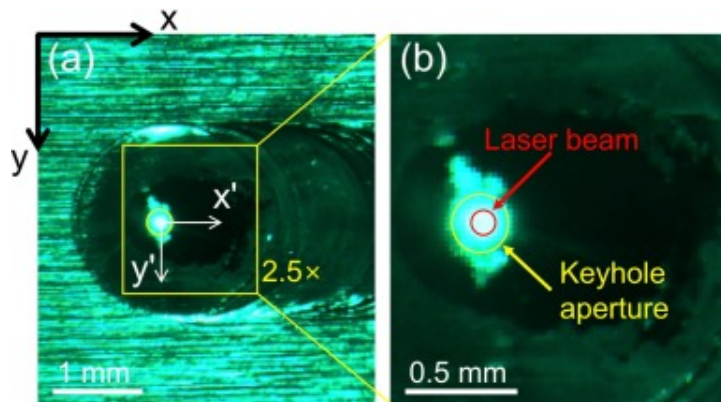
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Fig. 3. Laser power-scanning speed diagram showing 14 experimental conditions with the corresponding melt pool images. Red, black, and blue boundaries denote full, partial, and insufficient penetration results, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3. Dataset preparation

As shown in Fig. 1, our absorptance prediction scheme consists of three deep learning models, each of which requires a different dataset. This section explains the inputs to the model and the corresponding data labeling methods.

Fig. 4(a) shows a sample image used as input for both the object detection algorithm and the CNN regressor 1. As shown, the molten aluminum was found to be transparent, and the boundary between the transparent molten aluminum and the solid aluminum is clearly visible, as reported in our recent study [17]. Furthermore, information on the keyhole top aperture and the location of the laser beam can also be found in the figure. For better demonstration, a 2.5-times magnified image of the yellow box area in Fig. 4(a) is presented in Fig. 4(b). Note that the bright region shows thermal radiation, as the laser light was blocked by a band-pass filter [16]. In this figure, the keyhole top aperture and laser beam are marked as a yellow ellipse and a red circle, respectively. In this study, melt pool images were used as input without any image post-processing.



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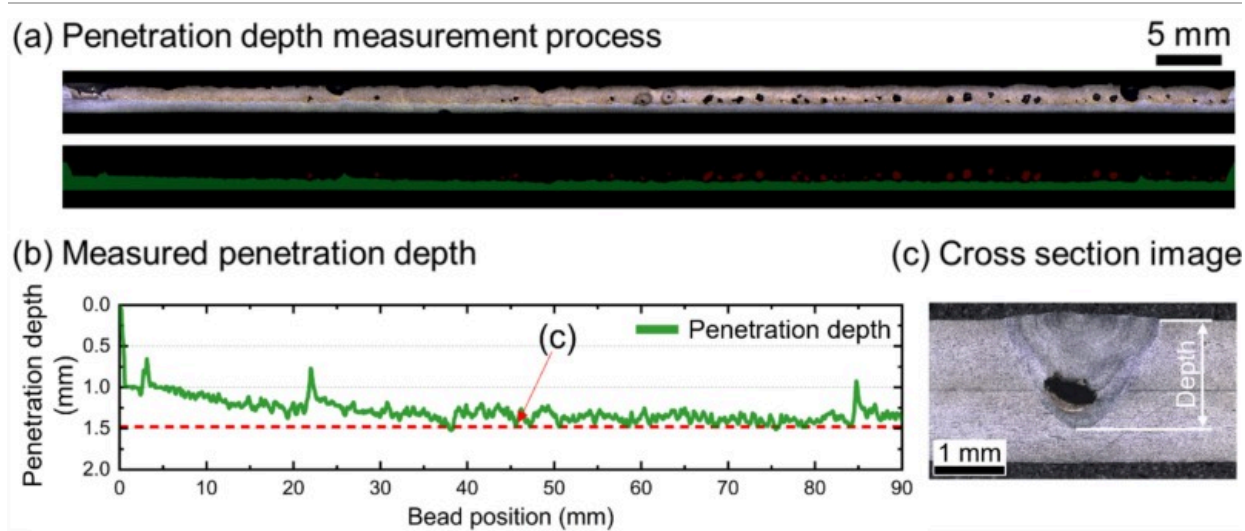
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Fig. 4. (a) A sample weld pool image used as input for both the object detection algorithm and the CNN regressor 1. (b) 2.5-times magnified image of the yellow box area in (a). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To train the object detection algorithm, we performed precise labeling by assuming that the keyhole aperture is elliptical, as depicted in Fig. 4(b). The keyhole aperture was labeled on a pixel-by-pixel basis using ImageJ software, and the aperture center

coordinates, as well as the lengths of the x and y axes, were measured. The pixel-based coordinate system (x–y coordinate system) is shown in the top-left corner of Fig. 4(a). Labeling was performed using this pixel coordinate system (352×320 resolution) with a spatial resolution of 11.7 μm per pixel. However, for the analysis of the results, a new coordinate system was constructed with the center of the laser beam as the origin, as shown in Fig. 4(a) (x'–y' coordinate system).

To train the CNN regressor model 1, we labeled the input melt pool images with the measured penetration depths corresponding to each image. Fig. 5 shows a data labeling procedure for the process condition with a laser power of 665W and a scanning speed of 11.2 mm/s. As shown at the top of Fig. 5(a), we cut and polished the welded specimen along the weld centerline in the welding direction. Then, the molten region (black) and the base metal region (green) were separated using LABELME software [18], as shown at the bottom of Fig. 5(a). In this way, we were able to obtain the penetration depths at all locations along the entire weld bead. The measurement results of Fig. 5(a) are presented in Fig. 5(b). To validate this measurement method, we performed an additional welding experiment using the same process condition and obtained a cross-sectional bead image (Fig. 5(c)) at a specific location. As shown in Fig. 5(b), the penetration depth obtained from the lateral cross-sectional image (red dashed line) is very close to the one obtained from the longitudinal image at the marked location.



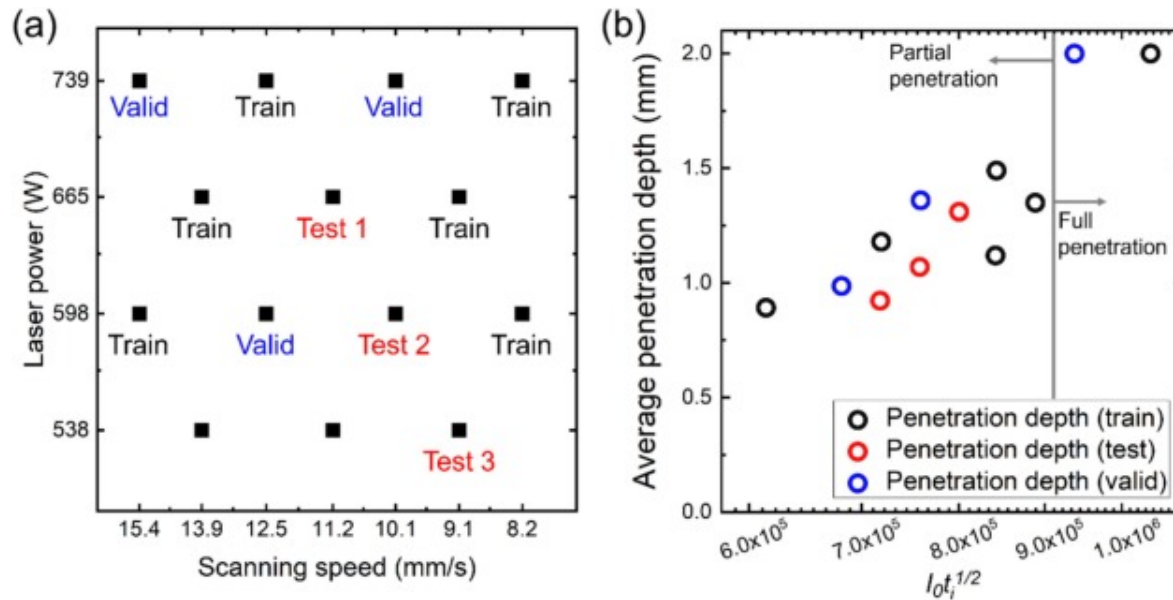
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Fig. 5. (a) Top: Cross-section of a welded specimen along the centerline in the welding direction and, Bottom: The molten region and base metal region are separated for penetration depth measurement. (b) The measured penetration depth versus the bead position. (c) Lateral cross-sectional image for measuring penetration depth. The measured depth is shown as a red dashed line in (b). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To prevent overfitting and ensure the generalization of the deep learning models, the two datasets prepared for the CNN regressor 1 and the object detection algorithm must be properly divided into training, validation, and test sets [19]. Fig. 6(a) shows all 14 experimental conditions on a laser power-scanning speed diagram. Out of the 14, the two conditions on the bottom left side were eliminated for insufficient penetration, and the remaining 12 were divided into 6 training, 3 validation, and 3 test sets. In Fig. 6(b), the average penetration depths for the 12 cases were plotted versus $I_0 t_i^{1/2}$, where I_0 and t_i represent the laser intensity (laser power/beam area) and interaction time (scanning speed/beam diameter) [20]. We can notice that all three datasets are well distributed in

terms of penetration depth. The dataset for the CNN regressor 1 used 35,478 training images, 14,393 validation images, and 18,960 test images. The object detection algorithm used 300 training images, 150 validation images, and 150 test images. From each process condition, 50 data points were randomly selected. In this study, full penetration conditions were employed to train CNN regressor 1 and the object detection model, both of which were used to obtain keyhole geometry. However, data from full penetration conditions were not used to train CNN regressor 2.



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Fig. 6. (a). Fourteen experimental conditions shown on a laser power-scanning speed diagram. (b) Average penetration depth plotted versus $I_0 t_i^{1/2}$ for the 12 conditions used for training, validation, and test datasets.

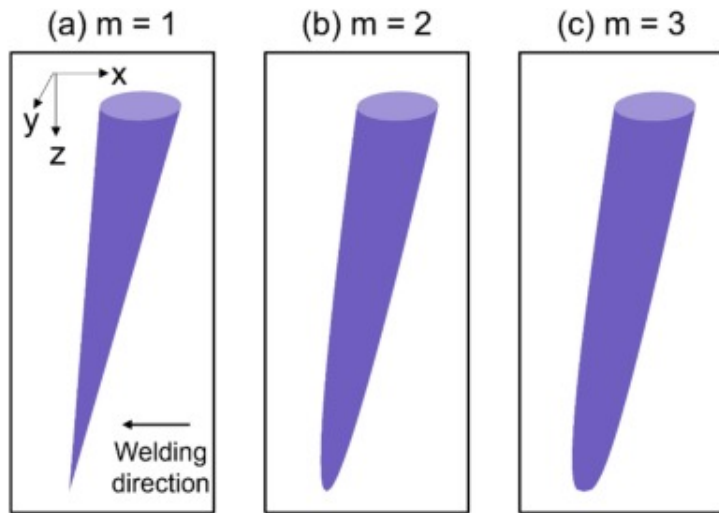
Once the top aperture shape and penetration depth were predicted from a melt pool image by the object detection model and CNN regressor 1, a keyhole shape was constructed as a two-channel image, which was the input to CNN regressor 2. Estimating

the shape of a partially penetrated keyhole is challenging, as the keyhole interior is not directly visible. Various assumptions have been made about the keyhole shape in many studies. Based on the generalized piston model, Fabbro [21] modeled the keyhole as a semi-cylindrical shape at low welding speeds, while at high welding speeds, the keyhole front wall was modeled as a half-cone shape. In one study, the keyhole was considered to have a cone-shaped and a stretched-hemisphere cavity [22]. In other studies [12,23], the keyhole shape was extracted from x-ray imaging. In this study, we used the following mathematical formula:

$$f(x, y, z) = \begin{cases} x = x_c - z \tan \theta + (1 - (z/d)^m) a \cos \varphi \\ y = y_c + (1 - (z/d)^m) b \sin \varphi \end{cases}, \quad (1)$$

based on the detected aperture center coordinates (x and y), x and y axis lengths of the elliptically assumed top aperture (a and b), and keyhole penetration depth (d). In Eq. (1), φ varies from 0 to 2π , θ is the keyhole tilting angle, and m is the degree of polynomial used to represent the keyhole surface. According to X-ray observation studies [3,24] and simulation studies [25], the tilting angle of a partially penetrated keyhole closely matches the laser head tilting angle. Consequently, a value of 10° was assigned to θ . Additionally, based on the studies utilizing optical coherence tomography [26] and X-ray observation [24], the melt pool depth was adopted as the keyhole penetration depth in this study.

While the aperture shape, penetration depth, and tilting angle have been determined, it remains necessary to establish an appropriate function that integrates all known parameters to model the keyhole as a three-dimensional surface. In this study, as illustrated in Eq. (1), a polynomial function with an exponent m was proposed. Fig. 7 depicts keyhole surfaces for m values of 1, 2, and 3, corresponding to a penetration depth of 1 mm. As shown, the keyhole forms a tilted cone when m=1 and becomes progressively rounder as m increases. In this study, various m values were tested for monitoring purposes, and m=2 was selected.

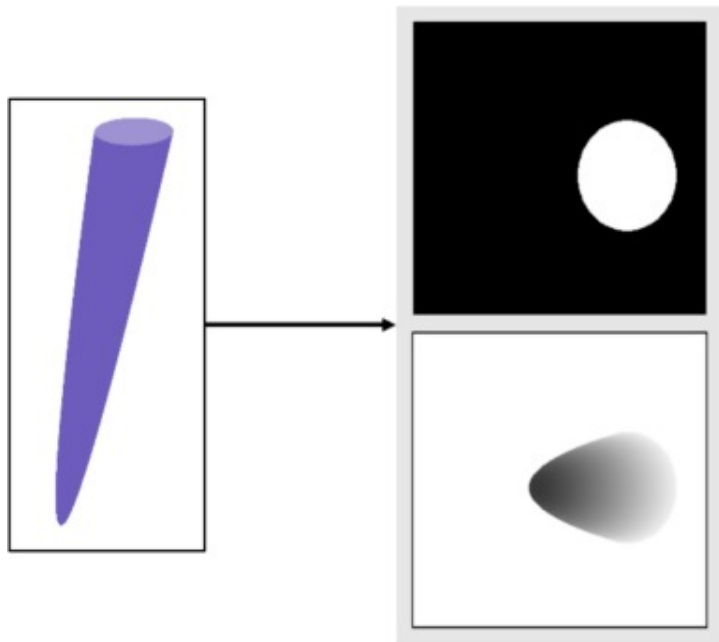


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Fig. 7. (a)-(c) Keyhole shapes according to Eq. (1) for m values of 1, 2, and 3, corresponding to a penetration depth of 1 mm.

For the input of CNN regressor 2, a two-channel keyhole image ($2 \times 288 \times 288$ pixels) was generated based on the keyhole equation (Eq. (1)). Fig. 8 illustrates an example of this two-channel keyhole image (right) alongside a corresponding three-dimensional keyhole (left). The first channel (top) shows the top aperture shape based on the detected aperture data. The second channel (bottom) visualizes the keyhole depth and shape using a grayscale, where 0 mm and 2 mm are mapped to 255 and 0, respectively, meaning that deeper areas appear darker. The dataset for CNN regressor 2 comprised 6000 images, including 4200 for training, 1200 for validation, and 600 for testing. It is noteworthy that the input images were generated using the data obtained from the object detection algorithm and CNN regressor 1 during actual laser welding experiments.



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Fig. 8. (Left): A 3-D keyhole constructed using Eq. (1) with the measured keyhole data. (Right) The 3-D keyhole is represented as a two-channel image, which is then used as input to the CNN regressor 2.

Using the keyhole equation (Eq. (1)) and for a wide variety of keyhole configurations, we performed ray tracing simulations to obtain keyhole absorptance values, which were used as the labels for the CNN regressor 2. To compute the absorptance values, a ray-tracing simulation was performed by discretizing a laser beam with a diameter of $100\mu\text{m}$ into a grid of $5\mu\text{m} \times 5\mu\text{m}$ beamlets. The ray tracing was conducted based on angle-dependent Fresnel equations [27]. Circular polarization was assumed, and its surface reflectance was calculated as an average of the reflectance values for p-polarized (R_p) and s-polarized (R_s) lights as

$$R = (R_p + R_s)/2, \quad (2)$$

where, R_p and R_s are

$$R_p = \frac{[n-1/\cos\theta_i]^2 + \kappa^2}{[n+1/\cos\theta_i]^2 + \kappa^2}, \text{ and} \quad (3)$$

$$R_s = \frac{[n-\cos\theta_i]^2 + \kappa^2}{[n+\cos\theta_i]^2 + \kappa^2}, \quad (4)$$

respectively. Here, θ_i represents the incident angle, while n and κ are the refractive index and extinction coefficient of aluminum, respectively, which are 1.75 and 8.5 [27]. Then, the local surface reflectance is given by

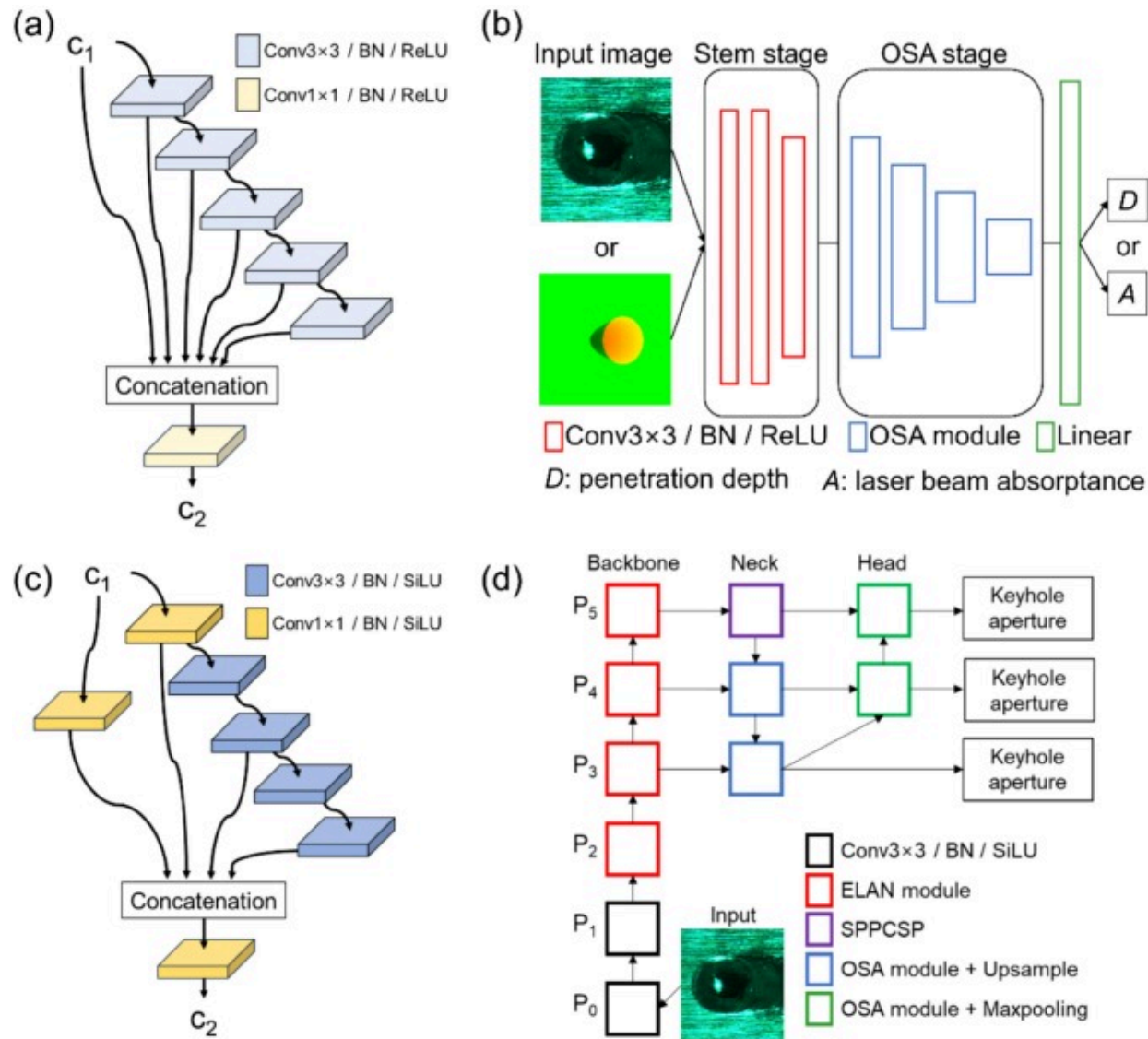
$$A = 1 - R. \quad (5)$$

A detailed explanation of the ray tracing algorithm can be found in [8].

From ray tracing simulations, it was observed that the absorptance distribution of the keyhole ranged from 9% to 80%. After normalizing this range to 0 to 1, 600 keyhole images were generated at 0.1 intervals to create a balanced dataset.

4. Deep learning models

The Variety of View Network (VoVNet) [15], a CNN architecture, was used for CNN regressors 1 and 2 to predict penetration depth and laser beam absorptance. VoVNet is known for its effective image feature extraction capabilities through the implementation of a one-shot aggressive (OSA) module, as shown in Fig. 9(a). It has been documented that the model outperforms established image feature extraction architectures like ResNet [28] and DenseNet [29] in terms of both accuracy and speed of inference [30]. In this study, we utilized the compact VoVNet27-slim model. Fig. 9(b) illustrates the structure of VoVNet27-slim, where a melt pool image or two-channel keyhole image is passed through the stem stage, OSA stage, and linear layer to predict the depth of penetration or absorptance.



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Fig. 9. (a) The OSA module used in VoVNet. (b) The VoVNet27-slim structure used for CNN regressors 1 and 2. (c) The ELAN module used in YOLOv7. (b) The structure of YOLOv7.

To predict a scalar value, we changed the output size of the final linear layer to one. This modification led to a shift from using the classifier loss function to employing the regressor loss function in the last layer of VoVNet27-slim. In this study, the mean absolute error (MAE) loss function was used to update the losses as follows:

$$L(x) = \frac{1}{N} \sum_{i=1}^N |W_i - \widehat{W}_i|, \quad (6)$$

where W_i represents the ground truth (measured penetration depth or absorptance simulated by ray tracing), and $\widehat{W}_i$ denotes the predicted values (predicted penetration depth or absorptance), and N is the number of data points.

For the object detection algorithm, YOLOv7 [14] was employed. YOLOv7 is a one-stage detection model that has been reported to be faster than two-stage detection models while maintaining comparable detection accuracy. The YOLOv7 model follows the YOLOR [31] baseline, so the model consists of a backbone, neck, and head, as shown in Fig. 9(d). The main change was that the backbone of the YOLOv7 architecture has been updated from YOLOR's cross-stage partial (CSP) block-based backbone to an efficient layer aggregation network (ELAN)-based backbone (Fig. 9(c)). ELAN was developed from VoVNet's OSA module by modifying the cross-stage connection and gradient path. Consequently, ELAN not only enhanced detection accuracy but also mitigated performance degradation as the model became deeper.

In this study, YOLOv7 was modified to perform single-class object detection by reducing the number of class channels in the detection head from 80 (as used in the COCO dataset) to 1, corresponding to the target class in this work. This modification simplified the detection head and reduced the number of trainable parameters, thereby improving training stability for our single-class dataset. The model was trained from scratch without using pretrained weights.

The object detection model was trained to minimize a loss function that combines complete IoU (CIoU) loss [32], objectness loss, and classification loss as follows:

$$L = \lambda_1 L_{CIoU} + \lambda_2 L_{objectness} + \lambda_3 L_{classification}, \quad (7)$$

where λ_1 , λ_2 , and λ_3 is 0.05, 0.7, and 0.3, respectively.

To train CNN regressors 1 and 2, as well as the object detection algorithm, we utilized the PyTorch deep learning library and employed four NVIDIA RTX A5000 GPUs with the distributed data parallel method. We used a trial-and-error approach to determine the optimal parameters. The detailed training parameters and results are summarized in [Table 2](#).

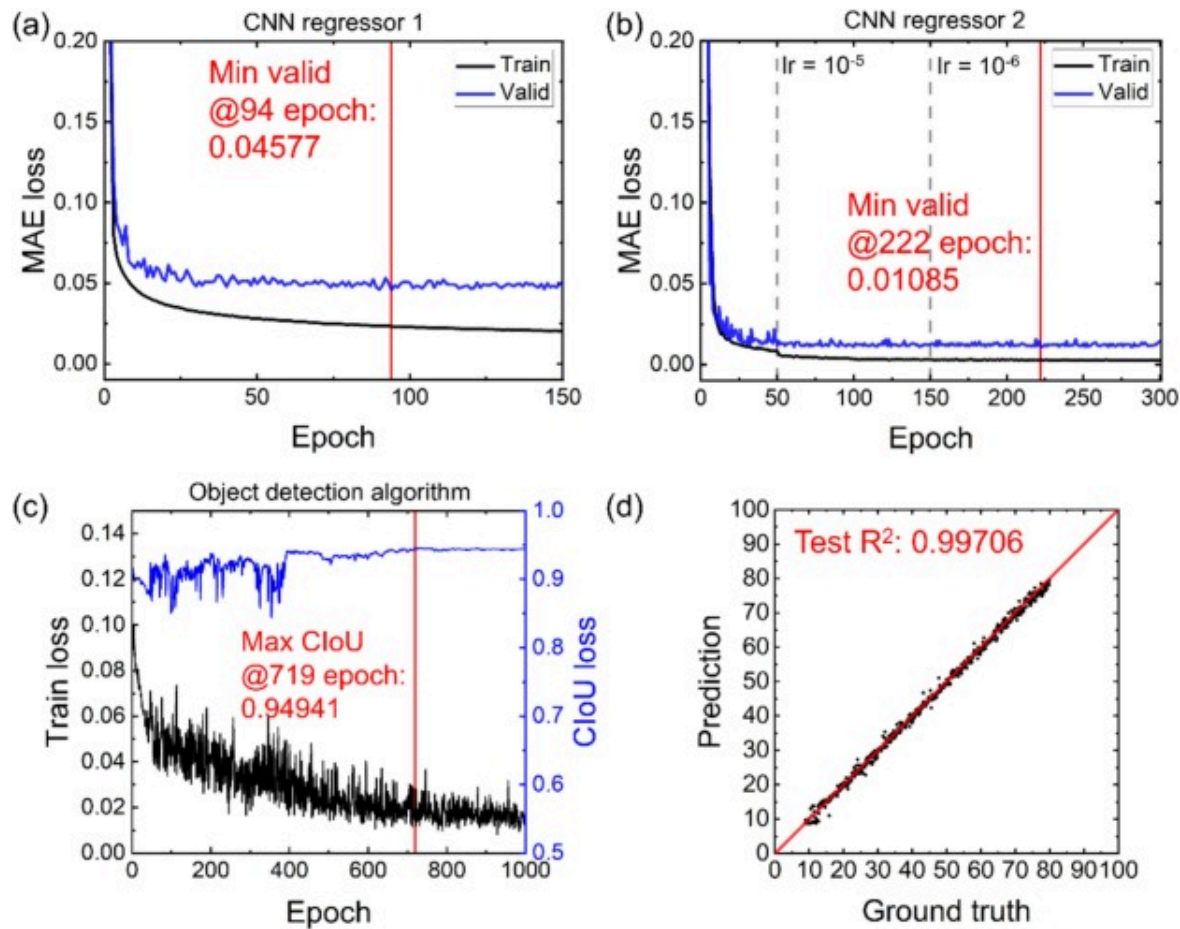
Table 2. Hyperparameters used for the CNN regressors 1 and 2 and the object detection algorithm.

Model	CNN regressor 1	CNN regressor 2	Object detection algorithm
Total epochs	150	300	1000
Batch size	64 (16 per 1 GPU)	40 (10 per 1 GPU)	128 (32 per 1 GPU)
Learning rate	10^{-6} to 5^{-6} (cosine decay method [33])	10^{-4} (decayed by 1/10 at 50 and 150 epoch)	10^{-2} to 10^{-4} (cosine decay method [33])
Weight decay coefficient	10^{-6}	10^{-6}	5×10^{-5}
Optimizer	Adam ($\beta_1=0.9$)	Adam ($\beta_1=0.9$)	SGD (momentum=0.937)
Augmentation	Flip y (real-time augmentation)	Flip y (pre-augmentation)	Mosaic [34], Color jitter [34], scaling, shifting, flip y, paste in [35]
Test MAE	0.03727	0.01147	
Test average IOU			0.9216
Prediction time	13.36s at batch size 16 per 1 GPU (18,960 test	3.7s at batch size 10 per 1 GPU (600 test	6.1/2.8/8.9ms (inference/NMS/total) at 32 batch

Model	CNN regressor 1	CNN regressor 2	Object detection algorithm
	images)	images)	size (150 test images with 1 GPU)

As mentioned earlier, the dataset for the object detection algorithm consisted of only 300 training samples, while the datasets for CNN regressors 1 and 2 were relatively larger—35,478 and 4200 samples, respectively. This limited dataset size may raise concerns about overfitting. As shown in [Table 2](#), a range of real-time data augmentation techniques were applied to the object detection model to mitigate overfitting and ensure stable training. Mosaic augmentation [34] merges one image with three randomly selected training images into a 2×2 composite, allowing the model to learn from a broader diversity of aperture shapes and melt pool appearances. Color jitter augmentation [34] randomly adjusts hue, saturation, and brightness to improve robustness against variations in lighting and exposure. Paste-in augmentation [35] inserts randomly selected image regions into a target image, enhancing the model's ability to detect objects in visually complex environments. Additional techniques—such as scaling, shifting, and vertical flipping (flip y)—were used to increase robustness to variations in aperture size, position, and orientation. These strategies were particularly effective for detecting small apertures, which are often difficult to capture. Collectively, these augmentations helped prevent overfitting despite the limited dataset size, contributing to the model's stable performance.

[Fig. 10](#) presents the training results of CNN regressors 1 and 2, as well as the object detection algorithm. [Fig. 10\(a\)](#) shows the MAE loss curves for the training (black line) and validation (blue line) data of CNN regressor 1. As training progressed, the training loss decreased, with the minimum validation MAE being 0.04577 after 94 epochs. The test MAE at the 94th epoch was 0.03727.



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Fig. 10. (a, b) MAE loss curves for CNN regressors 1 and 2, respectively. (c) CloU loss curves for the object detection algorithm. (d) Prediction accuracy of the absorptance prediction model (CNN regressor 2) for the test dataset. The R-squared accuracy was 0.99706. In (a)-(c), black and blue lines represent training and validation losses, respectively.

Fig. 10(b) shows the MAE loss curves for the training (black line) and validation (blue line) data of CNN regressor 2. The training MAE loss gradually decreased, with the minimum

validation MAE being 0.01085 after 222 epochs. The test MAE at the 222nd epoch was measured at 0.01147.

Fig. 10(c) shows the training loss (black line) and the validation CIoU loss (blue line) for the YOLOv7-based object detection algorithm. The optimal epoch for the model was determined using the CIoU loss function. As shown in Fig. 10(c), the optimal epoch of 719 was identified with a CIoU loss of 0.94941. The model achieved a detection performance of 0.9957 for mAP@0.5, with both precision and recall at 0.9333. Fig. 10(d) presents the ground truth versus prediction graph for the test dataset, showing an R-squared value of 0.99706.

In this study, we conducted an ablation study comparing VoVNet27-slim with two commonly used lightweight CNN architectures— MobileNetV3-small [36] and ResNet18 [28]—for both CNN Regressor 1 and CNN Regressor 2. All models were trained and tested under identical conditions using the same dataset.

As summarized in Table 3, VoVNet27-slim consistently outperformed the alternative architectures in terms of both prediction accuracy and inference speed. Specifically, for CNN Regressor 1, VoVNet27-slim achieved the lowest test MAE of 0.03727, compared to 0.04264 for ResNet18 and 0.03924 for MobileNetV3-small. For CNN Regressor 2, VoVNet27-slim yielded the highest R^2 value of 0.99706, outperforming ResNet18 (0.99691) and MobileNetV3-small (0.98613).

Table 3. Ablation study results for CNN regressors 1 and 2 with different CNN architectures.

Model	CNN architecture	Test MAE for regressor 1	Test prediction time (ms)
		Test R^2 for regressor 2	
CNN regressor 1	MobileNetV3-small	0.03924	1.3
	ResNet18	0.04264	1.1

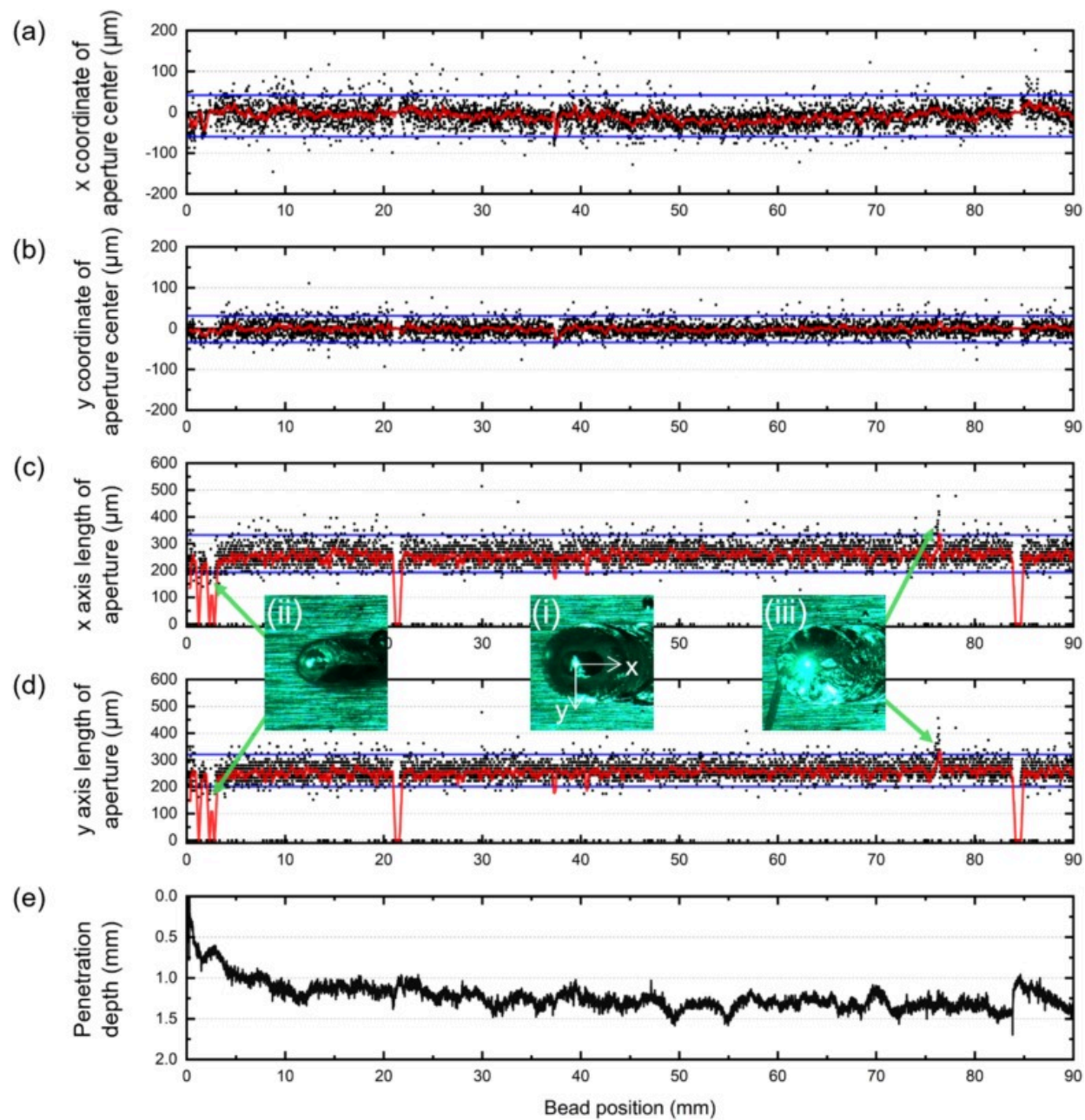
Model	CNN architecture	Test MAE for regressor 1	Test prediction time (ms)
		Test R ² for regressor 2	
CNN regressor 2	VoVNet27-slim	0.03727	0.7
	MobileNetV3-small	0.98613	21.3
	ResNet18	0.99691	19.8
	VoVNet27-slim	0.99706	6.2

The inference speed advantage of VoVNet27-slim was also significant. For CNN Regressor 1, VoVNet27-slim achieved a prediction time of 0.7ms, which was approximately 36% faster than ResNet18 (1.1ms) and 46% faster than MobileNetV3-small (1.3ms). For CNN Regressor 2, the improvement was even more pronounced: 6.2ms compared to 19.8ms for ResNet18 and 21.3ms for MobileNetV3-small—representing a more than threefold speed-up.

These results confirm the suitability of VoVNet27-slim for real-time absorptance monitoring in our application, as it offers both superior accuracy and computational efficiency.

5. Results and discussion

Fig. 11 illustrates the performance of CNN Regressor 1 and the object detection model in tracking keyhole aperture characteristics and predicting penetration depth during a laser welding process conducted at 665W and 11.2mm/s. Specifically, it presents the detected x- and y-coordinates of the keyhole aperture center (Fig. 11(a) and (b)), as well as the x- and y-axis lengths of the aperture (Fig. 11(a) and (b)). Additionally, Fig. 11(e) depicts the predicted penetration depth.



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Fig. 11. Performance of CNN regressor 1 and the object detection model in detecting keyhole aperture characteristics and predicting penetration depth during laser welding at 665W and 11.2mm/s. (a, b) Temporal variations of the keyhole's x- and y-coordinates. (c, d) Temporal variations of the x- and y-axis lengths, with red lines indicating time-averaged values (0.3mm) and blue lines showing two-standard-deviation ranges. (e) Predicted penetration depth along the bead position. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Unlike penetration depth, the aperture variables exhibit rapid and stochastic fluctuations. To mitigate noise and facilitate analysis, we computed their time-averaged values over a 0.3mm distance and highlighted them in red in Fig. 11(a)–(d). Furthermore, two-standard-deviation (SD) ranges are indicated with blue horizontal lines. Note that closed apertures were excluded when calculating the SDs for aperture lengths.

Fig. 11(a) and (b) illustrate the temporal variations of the keyhole's top aperture coordinates along the x- and y-axes, respectively, measured relative to the laser beam center. The x-coordinate (longitudinal) has a mean of $-8.98\mu\text{m}$ with an SD of $25.40\mu\text{m}$, whereas the y-coordinate (lateral) remains closely aligned with the welding centerline (mean: $-1.31\mu\text{m}$, SD: $16.27\mu\text{m}$). This suggests that the aperture exhibits greater displacement along the welding direction than in the lateral direction. Notably, the slightly negative mean x-coordinate indicates a minor shift in the welding direction, as shown in image (i) of Fig. 11.

Fig. 11(c) and (d) display the temporal variations of the x- and y-axis lengths of the keyhole aperture. The x-axis length has a mean of $262.67\mu\text{m}$ with an SD of $34.94\mu\text{m}$, while the y-axis length averages $260.90\mu\text{m}$ with an SD of $30.01\mu\text{m}$. These results indicate that the aperture is approximately circular, with slight elongation and more pronounced fluctuations in the welding direction.

The time-averaged aperture center coordinates and aperture lengths generally remain within the two-standard-deviation (SD) ranges. However, the time-averaged aperture

lengths occasionally fall outside these ranges, as observed in Cases (ii) and (iii) in Fig. 11. For comparison, Case (i) represents a stable melt pool with a regular aperture shape, corresponding to a stable keyhole.

Case (ii) illustrates an instance where the time-averaged aperture length drops below the lower 2 SD boundary. This reflects a moment when the aperture has not yet stabilized sufficiently to reach its typical size. Conversely, Case (iii) depicts a scenario where the time-averaged aperture length exceeds the upper 2 SD boundary, indicating a sudden shrinkage of the molten pool followed by a rapid expansion of the aperture. This suggests that the aperture size briefly increased within a very short period.

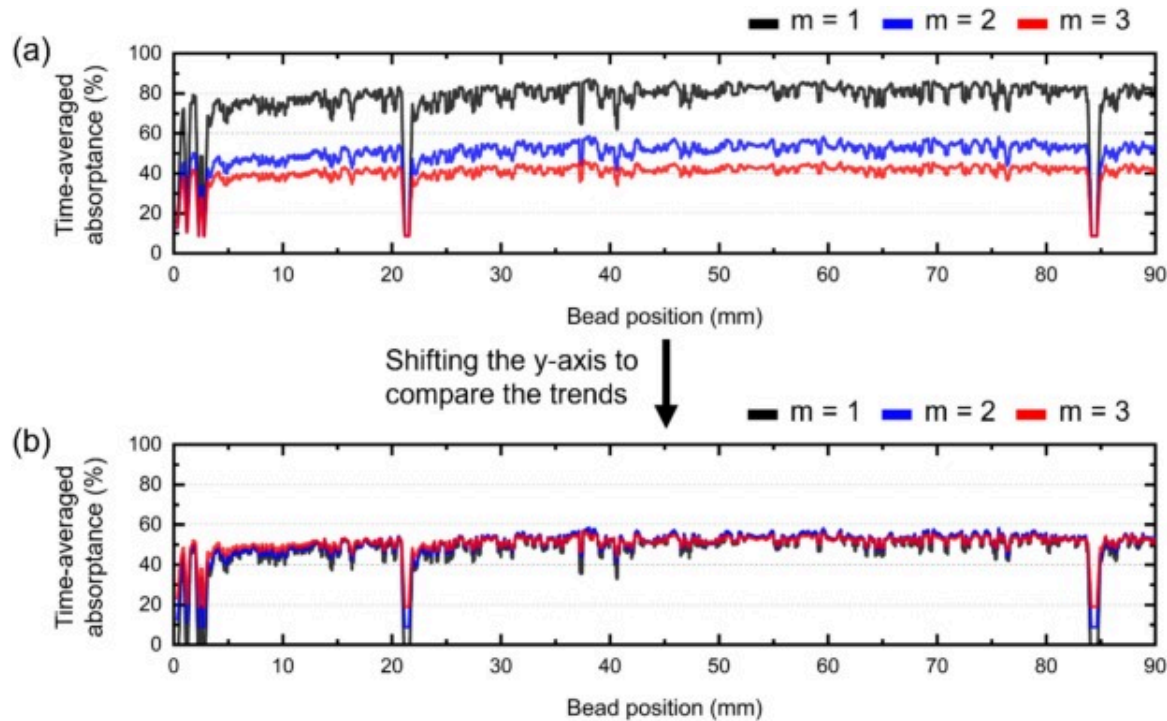
Fig. 11(e) shows the penetration depth throughout the welding process. As the keyhole depth increases, the number of multiple reflections generally rises, leading to higher absorptance [22]. As shown in the figure, the penetration depth gradually increases from 0mm to 1 mm up to a bead position of approximately 9mm. Beyond this point, it fluctuates between 1 mm and 1.5mm. These variations in keyhole depth influence changes in absorptance. Furthermore, as observed in Fig. 11(c) and (d), the keyhole momentarily closes at two positions (22.5mm, 85mm), where a sharp drop in keyhole absorption is expected.

Unlike full penetration welding, partial penetration makes it extremely difficult to observe the full geometry of the keyhole due to limited optical access to the lower region. Conventional in-situ monitoring techniques, such as inline coherent imaging (ICI) and optical coherence tomography (OCT), are capable of measuring penetration depth with high temporal resolution, but they typically provide only partial geometric information. Fleming et al. [12] employed ICI with a high acquisition rate of 280kHz to measure keyhole depth, and validated the results using synchrotron X-ray imaging. He et al. [13] utilized OCT operating at 80kHz for penetration depth measurement, combined with post-processing to align the OCT data with actual weld profiles. While these techniques provide high temporal resolution, they are primarily limited to depth measurement. In contrast, our approach uses a vision-based camera system combined with a deep learning

framework to estimate not only the penetration depth but also the laser beam absorptance, although it does not achieve the same temporal resolution as ICI or OCT.

In this study, CNN Regressor 1 achieved a test relative error of 6.77%. According to Fleming et al. [12], the ICI-based method produced depth measurements that matched X-ray imaging results with an approximate relative error of 6.22% for over 80% of evaluated samples. In the study by He et al. [13] using OCT, the application of a post-processing algorithm resulted in relative errors of 1.33% and 2.18% for two different test cases. While the prediction accuracy of our deep learning-based method is lower than that of OCT, it is comparable to the performance of ICI-based approaches. Considering that our method relies solely on vision-based image data without requiring specialized depth-sensing equipment, we believe this result demonstrates a promising level of accuracy for practical use in real-time process monitoring.

As discussed earlier, $m=2$ was selected to represent the keyhole for absorptance calculations. To validate this choice, absorptance was computed using ray-tracing simulations based on the results in Fig. 11, considering three values of $m=1, 2$, and 3. Fig. 12(a) presents time-averaged absorptance curves (computed over a 3mm interval) for $m=1$ (black), $m=2$ (blue), and $m=3$ (red). While all curves exhibit qualitatively similar trends, they differ in overall magnitude, with absorptance being highest for $m=1$ and decreasing as m increases. This occurs because a more rounded keyhole bottom (see Fig. 7) results in fewer multiple reflections of the laser beam.



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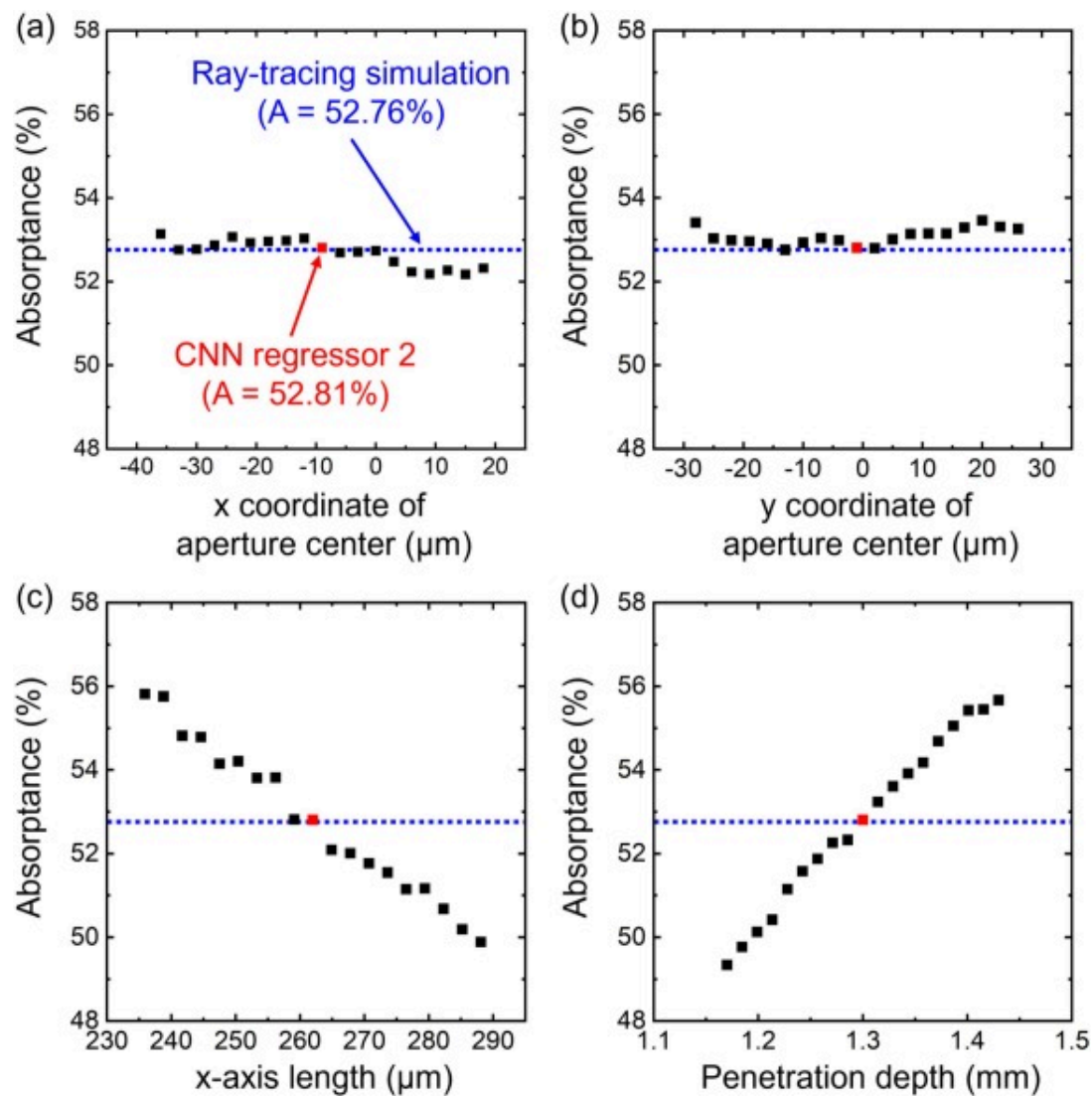
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Fig. 12. Time-averaged absorptance curves from ray-tracing simulations for $m=1$ (black), $m=2$ (blue), and $m=3$ (red). (a) Absorptance decreases as m increases but follows similar trends. (b) After vertical shifting, the curves overlap closely, supporting $m=2$ as a representative keyhole shape. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To further examine this trend, the absorptance curves for $m=1$ and $m=3$ were vertically shifted, revealing a strong overlap among all three curves, as shown in Fig. 12(b). This suggests that any of the three m values could be used for monitoring purposes, as only relative absorptance variations are important. This finding further supports the selection of $m=2$ as a representative keyhole shape in this study.

To investigate how prediction errors from the first-stage models (YOLOv7 and CNN Regressor 1) affect the absorptance prediction performance of the second-stage model (CNN Regressor 2), a parametric robustness study was conducted using a reference keyhole geometry. In this study, the input variables used to reconstruct the keyhole—namely the x- and y-coordinates of the aperture center, the x-axis length of the aperture, and the penetration depth—were systematically varied around their reference values to simulate potential prediction errors.

The results of this parametric study are presented in Fig. 13. As shown in Fig. 13(a) and (b), when the x- and y-coordinates of the aperture center were perturbed within $\pm 27\mu\text{m}$, the predicted absorptance remained close to the reference value, with maximum errors of 1.21% and 1.23%, respectively. These results indicate that CNN Regressor 2 is highly robust to positional deviations of the predicted aperture.



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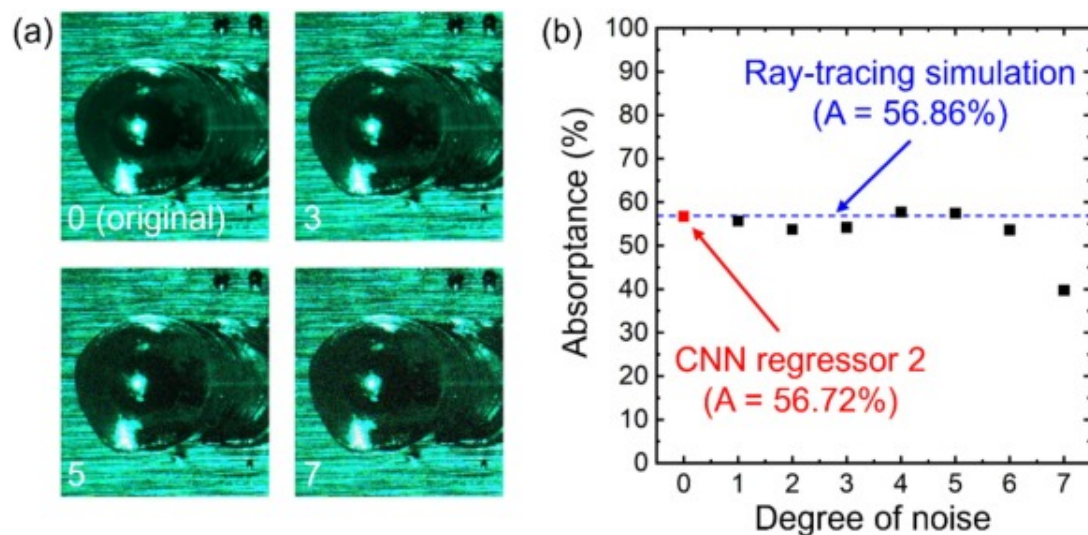
Fig. 13. Effect of input variations on absorptance prediction by CNN Regressor 2: (a) x-position, (b) y-position of aperture center, (c) x-axis length, and (d) penetration depth. Blue dashed lines: ray-tracing result; red square: CNN prediction for the reference

keyhole. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In contrast, variations in geometric parameters such as the x-axis length and penetration depth had a greater impact. As shown in Fig. 13(c), changing the x-axis length by up to $\pm 10\%$ resulted in a maximum prediction relative error of 5.70%. Similarly, varying the penetration depth within $\pm 10\%$ led to a maximum relative error of 6.57%, as shown in Fig. 13(d). Despite these larger variations, the prediction errors remained within a reasonably small range, demonstrating that the model can tolerate moderate input uncertainty while maintaining reliable absorptance predictions.

To further evaluate the robustness of the absorptance prediction framework under degraded input image conditions, we conducted an additional study using input images corrupted by increasing levels of random noise. In this experiment, random noise was generated using a Gaussian distribution with zero mean and standard deviation values ranging from 5 to 35 in steps of 5. These correspond to noise degrees 1 through 7, where a noise degree of n indicates a standard deviation of $5n$ (e.g., degree 7 corresponds to a standard deviation of 35). The noise filters were applied to the original image to create degraded inputs, while the noise-free input was labeled as degree 0.

Fig. 14(a) shows representative examples of the input images at different noise levels. As the noise degree increases, the visibility of keyhole features gradually deteriorates. Fig. 14(b) presents the predicted absorptance values corresponding to each noise level. The absorptance predicted by CNN Regressor 2 for the original image (degree 0) was 56.72%, which is in close agreement with the ray-tracing result of 56.86% (blue dashed line), yielding an error of only 0.14%.



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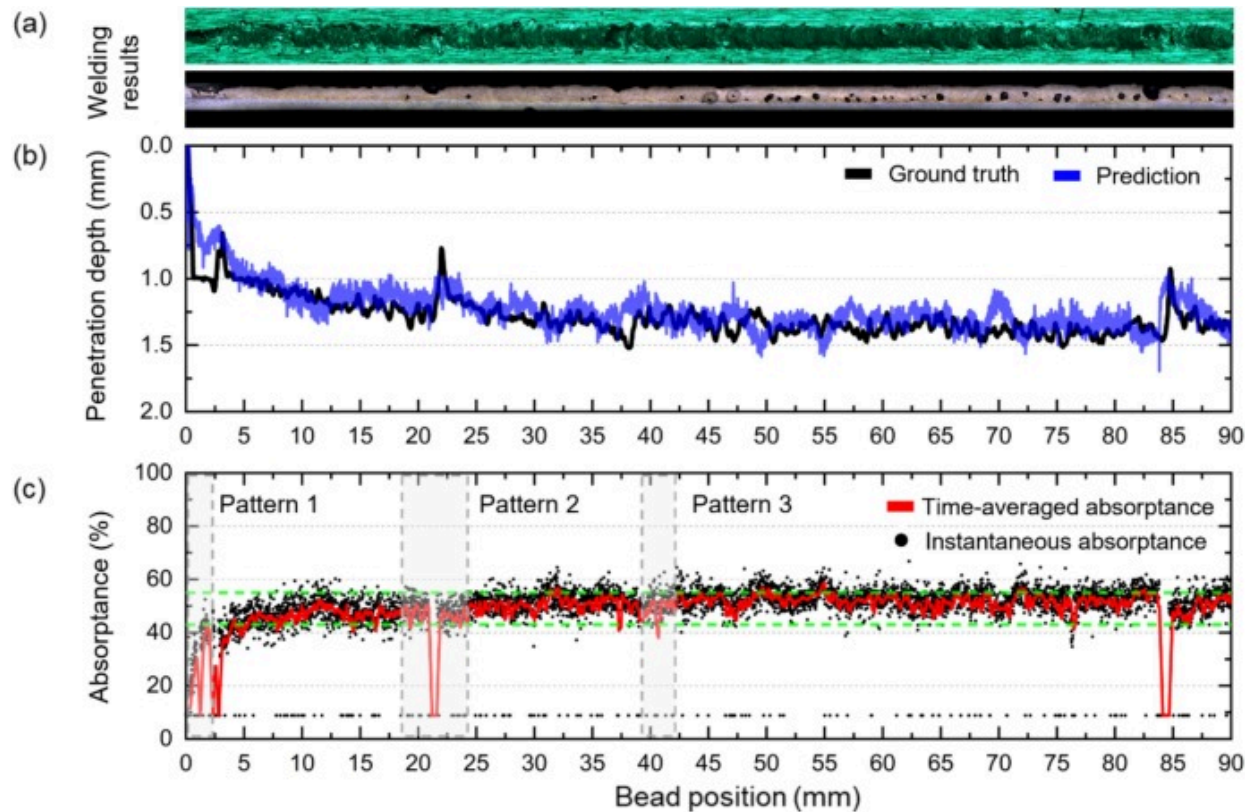
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Fig. 14. Effect of random noise on absorptance prediction by CNN Regressor 2. (a) Example input images at different noise levels. Noise degree n corresponds to Gaussian noise with zero mean and standard deviation of $5n^*$. (b) Predicted absorptance values for each noise level. Blue dashed line: ray-tracing result (56.86%); red square: prediction for noise-free image (56.72%). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Up to noise degree 6, the prediction error remained within approximately 5%, indicating that the model retains high robustness under moderate image degradation. However, at the highest noise level (degree 7), the prediction error increased significantly to about 30%, suggesting that the model's performance deteriorates under severe visual noise.

These results demonstrate that the proposed framework is capable of maintaining accurate absorptance predictions under a realistic range of image degradation, further confirming its robustness for use in practical, potentially noisy environments.

In this study, we tested the developed absorptance monitoring algorithm using three test cases shown in Fig. 6, with absorptance predictions based on the model outlined in Fig. 1. Fig. 15 presents the absorptance monitoring results from the first test experiment, conducted at a laser power of 665W and a scanning speed of 11.2mm/s, representing the highest energy density among the three test cases.



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Fig. 15. Absorptance monitoring results from the first test experiment (665W, 11.2mm/s). (a) Top view of the weld bead and longitudinal cross-section. (b) Measured (black) and predicted (blue) penetration depth. (c) Instantaneous (black dots) and time-averaged (red line) absorptance, with the stable range (43%–55%, green dashed lines). Three absorptance patterns were identified: (1) initial increase (left gray box), (2) prolonged

keyhole closure (middle gray box), and (3) temporary absorptance dips (right gray box). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 15(a) displays the top view of the weld bead and the longitudinal cross-section over the full 90mm weld length. Fig. 15(b) compares the measured penetration depth (black line) with the predicted penetration depth (blue line) as a function of bead position, which corresponds to the positions in Fig. 15(a). Fig. 15(c) presents the instantaneous absorptance distribution (black dots) and time-averaged absorptance (red line).

As shown in Fig. 15(b), penetration depth steadily increases until the bead position reaches 9mm, after which it fluctuates between 1.0mm and 1.5mm. At bead positions of 22.5mm and 85mm, the keyhole collapses, causing penetration depth to drop to zero. Additionally, at 3mm, shortly after welding begins, penetration initially deepens before suddenly becoming shallower. The penetration depth predictions from CNN Regressor 1 closely follow the trend of the measured values.

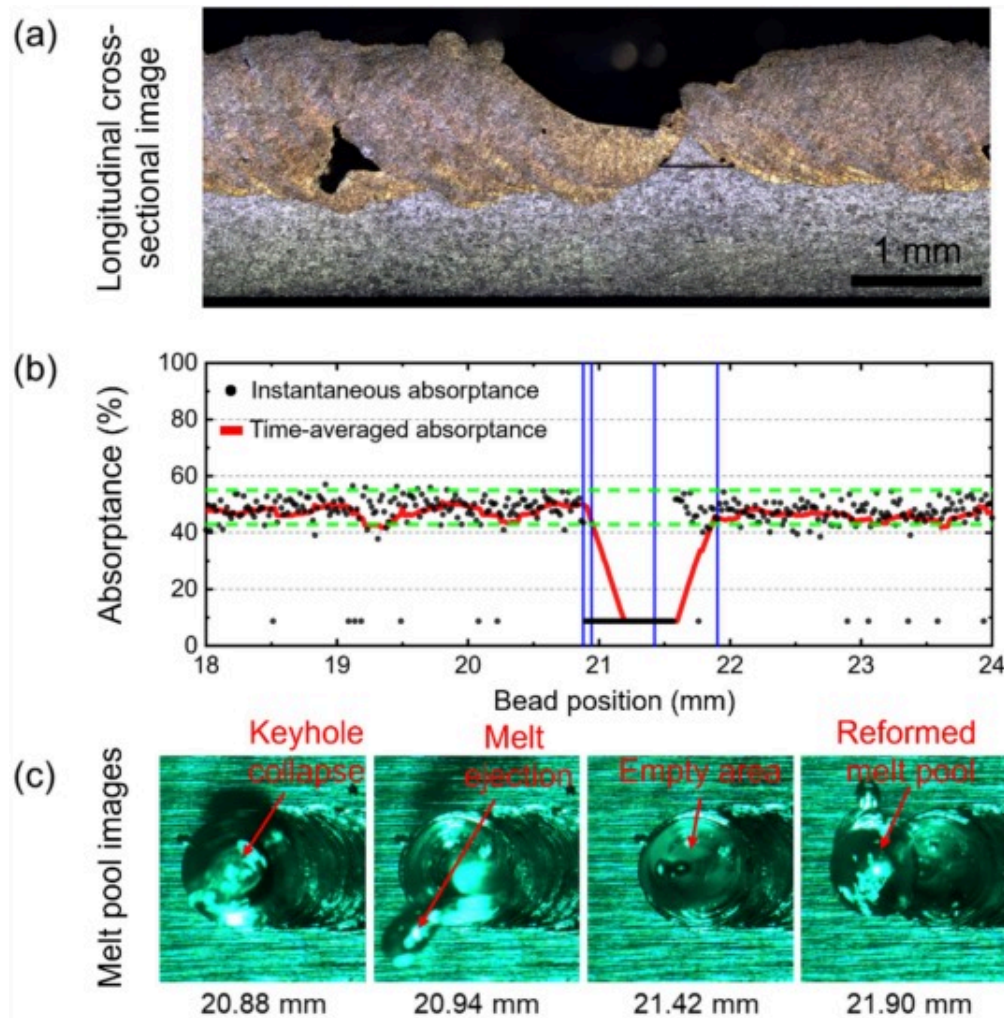
In Fig. 15(c), the instantaneous absorptance exhibits significant fluctuations, while the time-averaged absorptance follows a more stable trend. The time-averaged absorptance consistently falls within 43% to 55%, as indicated by the green dashed lines. This range is determined as the mean absorptance (49%) $\pm$ one standard deviation (6%). A stable keyhole is generally maintained within this absorptance range, while deviations outside it indicate unstable keyhole behavior, leading to changes in the absorption mechanism and a decline in welding quality.

To assess weldability, we categorized unstable absorptance conditions into three distinct patterns. The first absorptance pattern, shown in the left gray box in Fig. 13(c), occurs when absorptance gradually increases from 9% to 43%. Initially, the low absorptance results in insufficient penetration depth relative to the expected level for the given laser energy density. However, as absorptance increases, sufficient energy absorption is achieved, allowing penetration depth to reach the expected level. This transient phase

occurs briefly at the start of welding, during which the top bead gradually widens, as seen in Fig. 15(a).

The second absorptance pattern occurs when the keyhole remains closed for an extended period. One instance of this pattern is highlighted in the middle gray box in Fig. 15(c), where the time-averaged absorptance drops significantly to 9%, well below the typical range of 43% to 55%. This pattern appears at bead positions of 3 mm, 22.5 mm, and 85 mm in Fig. 15(c). Under these conditions, welding proceeds with very low absorptance, leading to a shallower penetration depth and a narrower top bead width than expected for the given energy input. Additionally, this pattern is associated with underfill defects.

Fig. 16(a) presents the longitudinal cross-sectional bead image corresponding to bead positions between 18 mm and 24 mm (middle gray box). Fig. 16(b) shows the predicted absorptance at the same bead positions along the weld line. As shown, the time-averaged absorptance drops to 9% for approximately 0.6 mm from the bead position of 20.88 mm, coinciding with keyhole collapse. This collapse is evidenced by the suppressed surface profile (underfill defect) and shallow penetration depth observed in the cross-sectional image.



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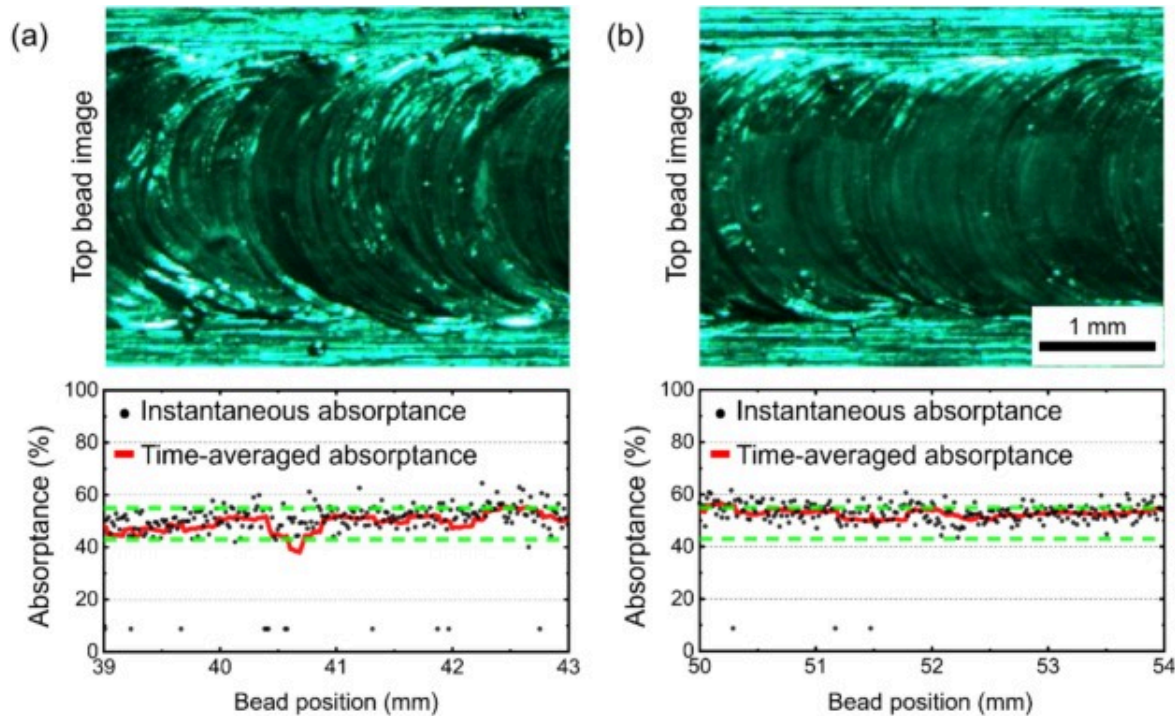
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Fig. 16. Keyhole collapse and its effect on absorptance and weld quality. (a) Longitudinal cross-section of the bead (18–24mm), showing an underfill defect with shallow penetration. (b) Predicted absorptance, dropping to 9% for 0.6mm from 20.88mm, coinciding with keyhole collapse. (c) Melt pool images at key positions: keyhole collapse

(20.88mm), melt ejection (20.94mm), empty area (21.42mm), and reformed melt pool (21.90mm) with a narrowed top bead.

Fig. 16(c) presents a series of melt pool images in this region. The sequence shows keyhole collapse, followed by melt ejection, which causes the molten material to disperse, preventing keyhole formation and limiting multiple reflections. This second absorptance pattern leads to reduced laser energy absorption, resulting in shallower penetration depth. Additionally, as seen at 21.90mm in Fig. 16(c), the top bead width initially narrows as absorptance decreases before recovering to its original width.

The third absorptance pattern (right gray box in Fig. 15(c)) is characterized by a temporary decrease in time-averaged absorptance, slightly dropping below the lower boundary of the one-SD range. Fig. 17(a) presents a top bead image and absorptance data at the corresponding bead positions. As absorptance decreases, the melt pool becomes unstable, leading to an irregular bead shape. Unlike the second absorptance pattern, absorptance does not drop drastically to 9%, preventing extreme changes in penetration depth. Instead, it primarily affects the top bead shape.



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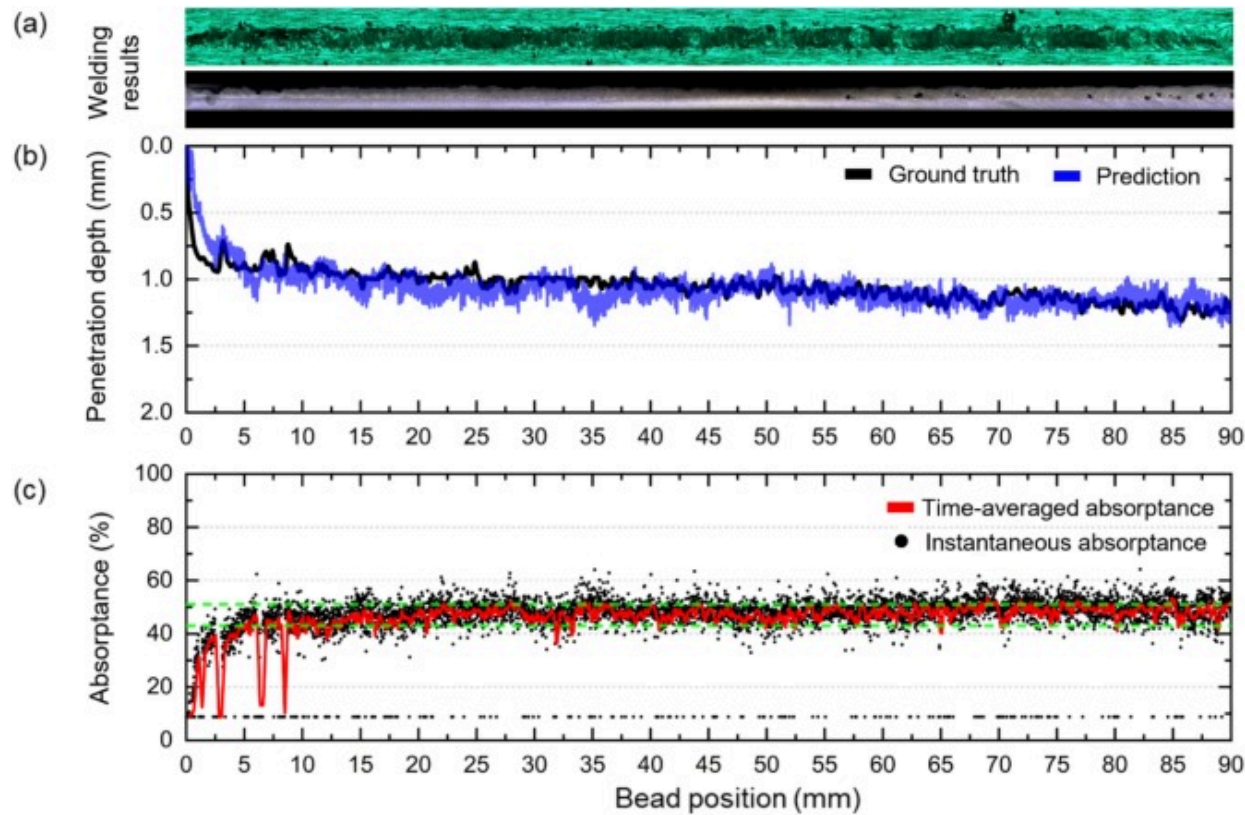
Fig. 17. Effect of absorptance fluctuations on top bead shape. (a) Top bead image and absorptance data for the third absorptance pattern, where a temporary drop in absorptance leads to an irregular bead shape. (b) Comparison with a stable bead shape (50–54mm), where absorptance remains nearly constant, resulting in regular ripple patterns indicative of high welding quality.

For comparison, Fig. 17(b) shows a stable bead shape at bead positions 50–54mm, where absorptance remains nearly constant. This region exhibits the most stable time-averaged absorptance, contrasting sharply with Fig. 17(a). Highly stable absorptance indicates a consistent welding process, resulting in regular ripple patterns on the top bead surface, a key indicator of high-quality weld appearance.

It is noteworthy that although keyhole geometry may change dynamically during laser welding, it is widely accepted in the laser welding community that energy absorption due to multiple reflections of the laser beam inside the keyhole is fundamentally an optical phenomenon that occurs on a much shorter timescale than the morphological evolution of the keyhole itself. Therefore, assuming that absorptance is determined by the instantaneous (static) keyhole shape is considered physically reasonable.

This assumption has been almost universally adopted in previous laser keyhole welding studies. In our study, the absorptance values calculated under this static geometry assumption were shown to correlate well with key process indicators—supporting the practical validity of this modeling approach.

[Fig. 18](#) presents the absorptance monitoring results for the second experiment (laser power: 598W, scanning speed: 10.1 mm/s). [Fig. 18\(a\)](#) displays the top weld bead and longitudinal cross-section over the 90mm weld length, while [Fig. 18\(b\)](#) compares the measured (black) and predicted (blue) penetration depths along the weld line. [Fig. 18\(c\)](#) illustrates the instantaneous absorptance distribution (black scatter) and time-averaged absorptance (red line).



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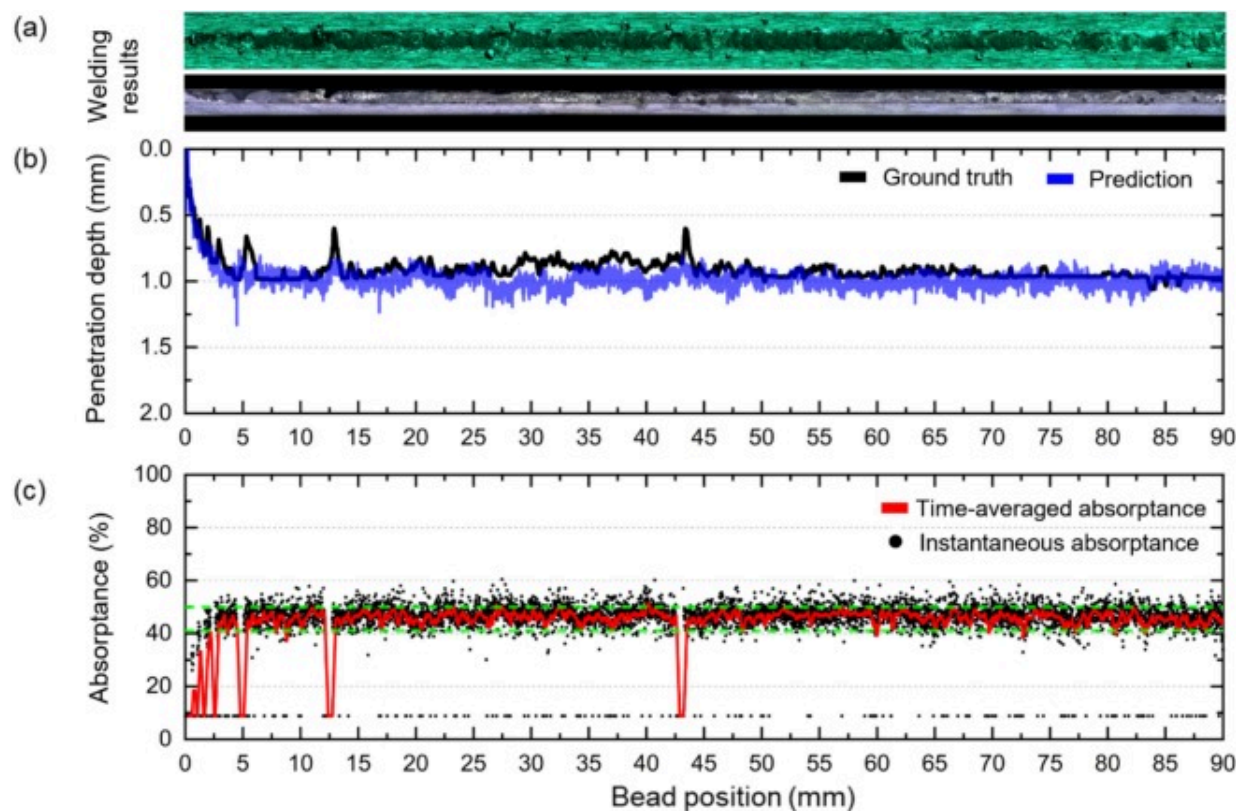
Fig. 18. Absorptance monitoring results for the second experiment (598W, 10.1 mm/s). (a) Top weld bead and longitudinal cross-section. (b) Measured (black) and predicted (blue) penetration depth. (c) Instantaneous (black dots) and time-averaged (red line) absorptance, with the stable range (43%–51%, green dashed lines). Three absorptance patterns: (1) initial increase (0–2.5mm) deepening penetration, (2) keyhole closure at 3mm, 6.5mm, 8.5mm causing underfill defects, and (3) temporary dips (34mm, 65mm) affecting the top bead shape. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

As in the first experiment, all three absorptance patterns were observed. In the first pattern, during the initial welding phase (0–2.5 mm), the time-averaged absorptance increases from 9% to 40%, leading to a gradual widening of the top bead width and deeper penetration depth.

Throughout most of the weld, time-averaged absorptance remained between 43% and 51% (mean: 47%, SD: 4%), as indicated by the green dashed lines in Fig. 18(c). Sudden absorptance drops occurred at bead positions 3 mm, 6.5 mm, and 8.5 mm, corresponding to the second absorptance pattern, in which the keyhole remained closed for a prolonged period. This resulted in shallower penetration depth, narrower top bead width, and underfill defects, as seen in Figs. 18(a) and 16(b).

The third absorptance pattern was also observed. For instance, at 34 mm and 65 mm, time-averaged absorptance showed a temporary decrease rather than dropping to 9% as in the second pattern. During these moments, the top bead shape became irregular, indicating poor weld quality. In contrast, at 35–39 mm, absorptance remained stable, producing a consistent ripple pattern in the top bead.

Fig. 19 presents the absorptance monitoring results for the third experiment (laser power: 538 W, scanning speed: 9.1 mm/s), which had the lowest energy density among the three test cases. Fig. 19(a) shows the top weld bead and longitudinal cross-section, while Fig. 19(b) compares the measured (black) and predicted (blue) penetration depths over the 90 mm weld length. Fig. 19(c) displays the instantaneous absorptance distribution (black scatter) and time-averaged absorptance (red line).



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Fig. 19. Absorptance monitoring results for the third experiment (538W, 9.1 mm/s). (a) Top weld bead and longitudinal cross-section. (b) Measured (black) and predicted (blue) penetration depth. (c) Instantaneous (black dots) and time-averaged (red line) absorptance, with the stable range (40%–50%, green dashed lines). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In the third experiment, both the first and second absorptance patterns were observed. During the initial welding stage, the first pattern emerged, where absorptance gradually increased until a stable keyhole was established. In the second pattern, absorptance

momentarily dropped from the typical range (40%–50%, mean: 45%, SD: 5%) to 9% at bead positions 5 mm, 12.5 mm, and 43 mm, leading to abrupt penetration depth reductions and underfill defects, as shown in Fig. 19(a) and (b). The third absorptance pattern was also present but was less pronounced due to the small fluctuation range of time-averaged absorptance, making it difficult to distinguish from normal variations. However, in the 50–54 mm section, as shown in Fig. 19(c), absorptance remained stable, producing a consistent top bead shape with a well-defined ripple pattern, as seen in Fig. 19 (a).

One thing to note is that laser welding of aluminum alloys is inherently unstable, and pores were observed in the welded specimens under many process conditions. To investigate whether our model is capable of predicting pore formation based on variations in laser beam absorptance, we examined our experimental dataset, comparing the timing and location of porosity formation with the corresponding time-averaged absorptance trends. However, no consistent correlation was found between the occurrence of porosity and deviations in the absorptance signal. Based on this observation, we conclude that while the current model effectively captures absorptance behavior, it does not yet have the capability to predict porosity formation.

To assess the practical feasibility of real-time implementation, we evaluated the computational performance of the proposed model on a workstation equipped with a single NVIDIA RTX A5000 GPU and an Intel Xeon Gold 6330 CPU. The latency of each model component was measured as follows: 10 ms for CNN Regressor 1 (penetration depth prediction), 13 ms for the object detection algorithm (YOLOv7), and 18 ms for CNN Regressor 2 (absorptance prediction), resulting in a total end-to-end latency of 31 ms per image. This corresponds to a processing speed of approximately 32.2 frames per second (fps).

In a practical application test, the system was operated at around 30 fps and successfully detected process transitions and defect formation in real time during actual welding. These results confirm that, even with standard high-performance hardware, the proposed

deep learning-based absorptance monitoring framework is capable of real-time operation and is suitable for industrial deployment.

6. Conclusions

The key findings of this study are summarized as follows:

- A deep learning-based absorptance monitoring system was developed for partial penetration laser keyhole welding of Al 1050P-H16. The system integrates YOLOv7-based object detection and two CNN regressors (VoVNet27-slim) to predict keyhole aperture shape, penetration depth, and laser absorptance from melt pool images.
- The trained models demonstrated high prediction accuracy, with penetration depth and absorptance prediction errors (MAE) of 0.03727 and 0.01147, respectively. The system successfully detected underfill defects and abrupt penetration depth changes, validating its effectiveness in real-time process monitoring.
- Keyhole dynamics and absorptance fluctuations were analyzed, revealing that keyhole openness and depth varied significantly throughout the process. Despite rapid stochastic fluctuations, the time-averaged absorptance exhibited a stable trend, providing a reliable metric for monitoring.
- Absorptance trends remained consistent regardless of the keyhole shape modeling parameter (m). When absorptance curves were shifted vertically, they closely overlapped, indicating that relative absorptance changes, rather than absolute values, are key to process monitoring. This finding confirms that absorptance monitoring is robust to variations in assumed keyhole shape.

- Three distinct absorptance patterns were identified, correlating with weld bead shape variations, penetration depth fluctuations, and defect formation. These patterns demonstrated that sudden absorptance drops were linked to keyhole collapse, penetration reduction, and underfill defects, highlighting the importance of absorptance stability in welding quality.
- This study presents the first deep learning-based absorptance monitoring system for partial penetration laser welding, providing valuable insights into hidden keyhole dynamics.

CRediT authorship contribution statement

Kimoon Nam: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis. **Hyungson Ki:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) ([RS-2024-00334308](#)).

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